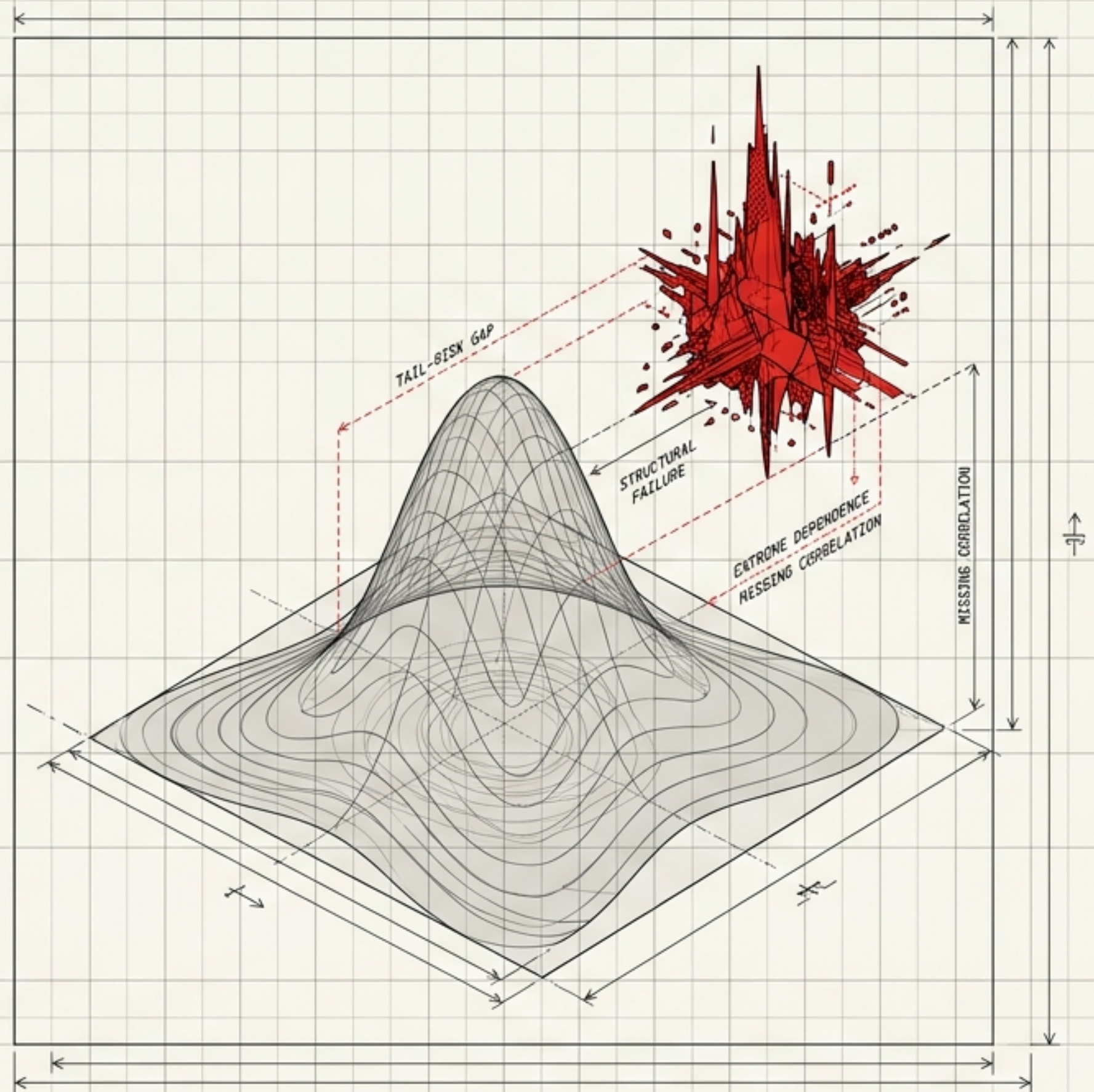


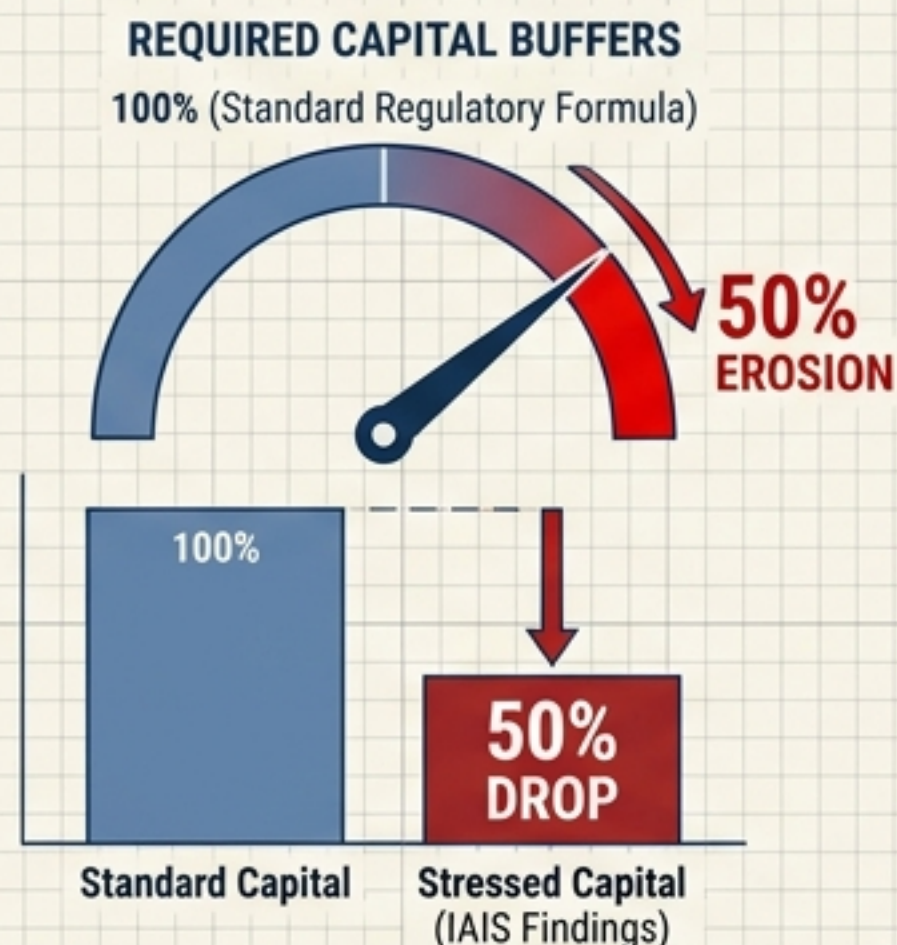
THE STRUCTURAL ILLUSION OF CLIMATE CAPITAL

Diagnosing the mathematical failure of actuarial correlation models—and the soft-weighted classifier that closes the tail-risk gap.



THE ACTUARIAL BLUEPRINT

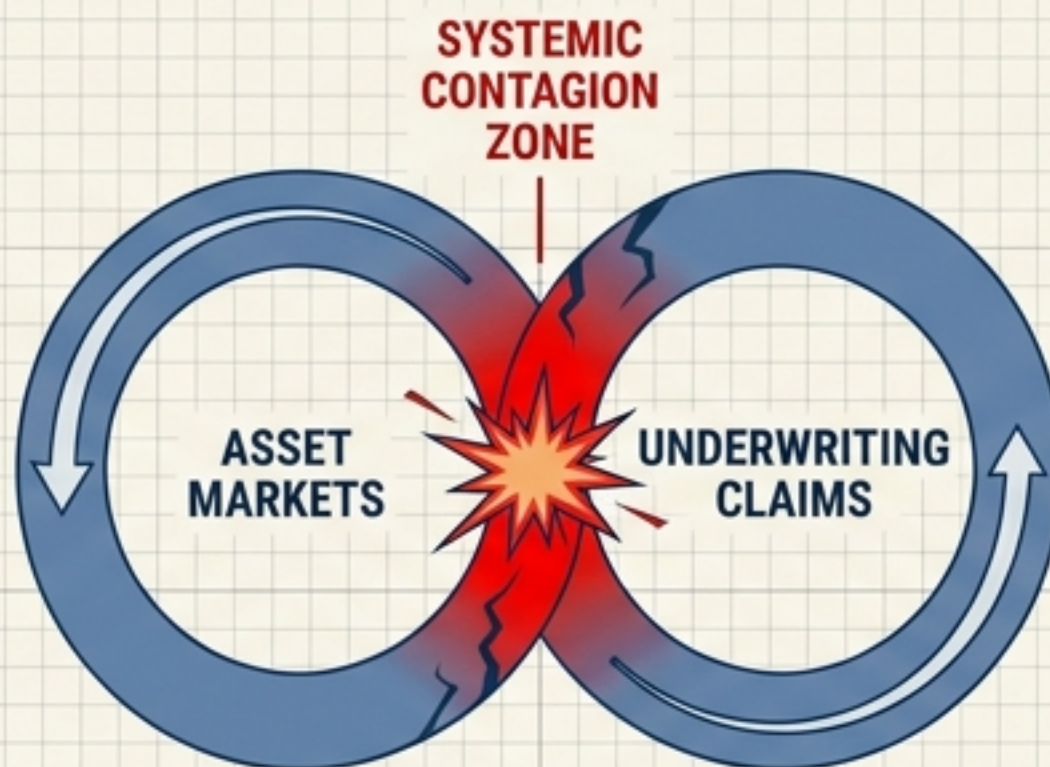
THE 50% CAPITAL EROSION



Regulatory Reality: Under non-linear physical/transition climate stress, standard regulatory formulas mask a potential **50% DROP** in required capital buffers.

Source: IAIS Global Insurance Market Report

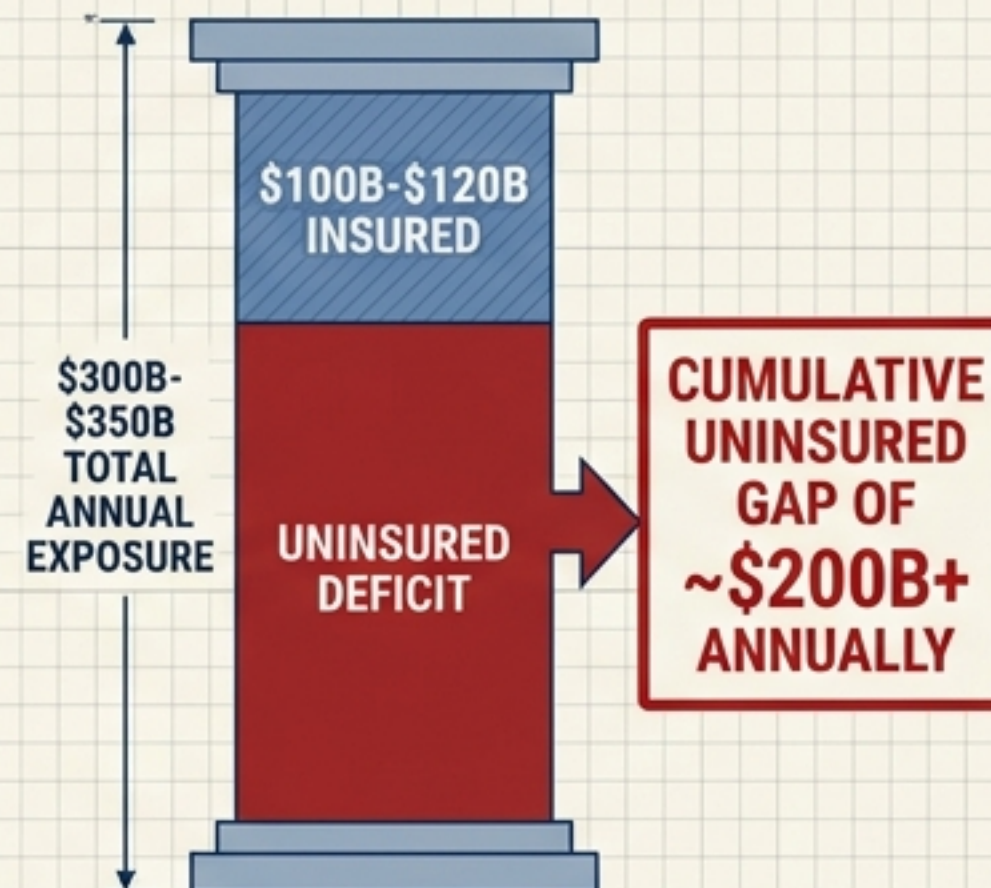
CRISK (CAPITAL SHORTFALL)



Systemic Contagion: Federal Reserve metrics show instantaneous capital shortfalls in the tens to **HUNDREDS OF BILLIONS**. Asset markets and underwriting claims degrade simultaneously.

Source: Federal Reserve Financial Stability Report

THE \$1.8 TRILLION PROTECTION GAP



The Uninsured Deficit: A cumulative uninsured gap of **~\$200B+** annually, forcing market collapses and state-backed residual pools.

Source: Various Industry Reports & Swiss Re Institute

THE INDUSTRY RELIES ON **THIN-TAILED** STATISTICAL ASSUMPTIONS THAT MASK **HUNDREDS OF BILLIONS** IN EXPOSURE.

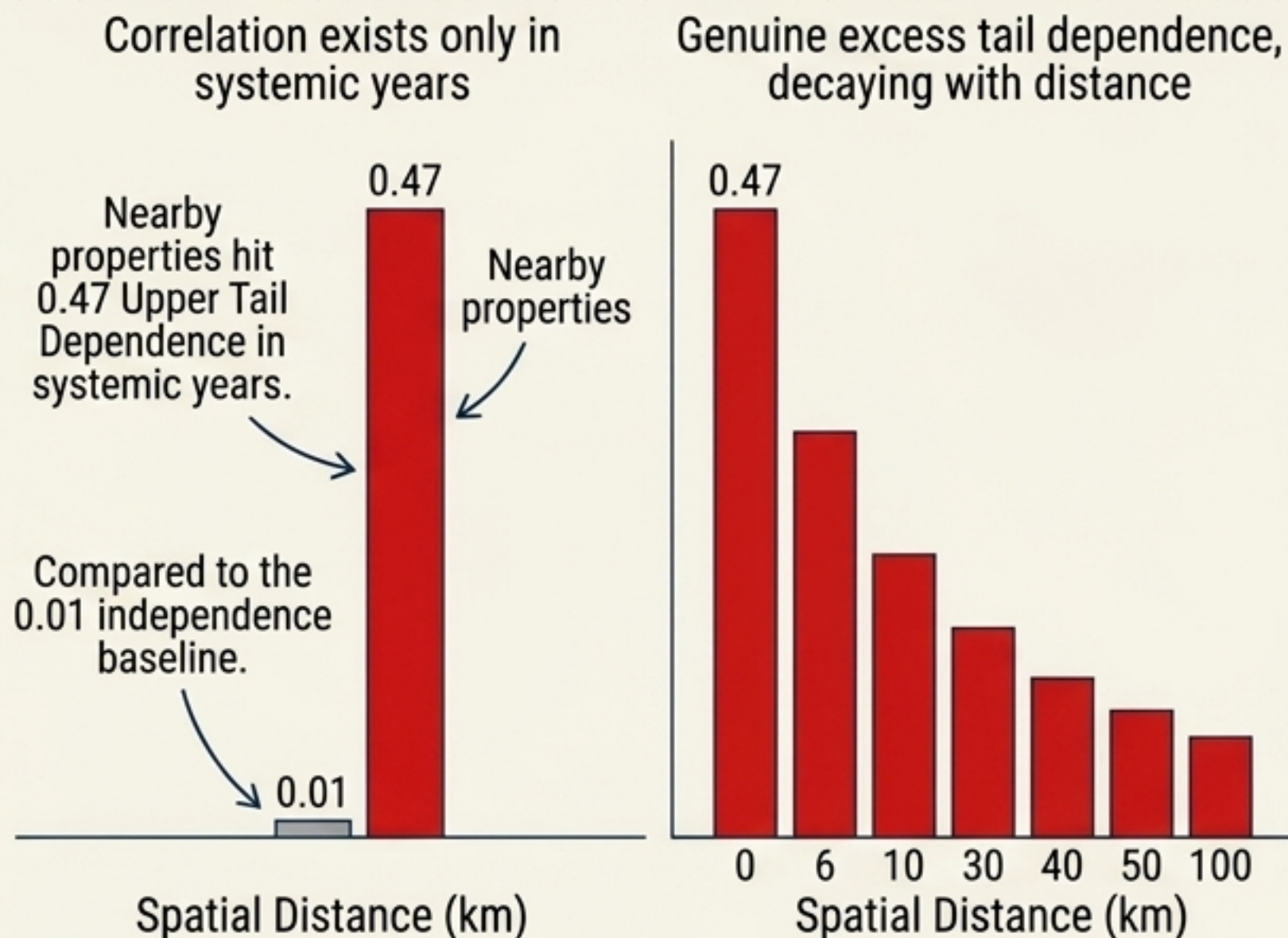
THE ACTUARIAL BLUEPRINT

THE 99.5% TARGET

$$C^* = \inf\{c : P(L \leq c) \geq 0.995\}$$

The insurer must hold capital C^* so that total losses L don't exceed it with 99.5% probability (the Solvency II 1-in-200-year convention).

THE TAIL DEPENDENCE PROOF



Key Insight: Gaussian-copula models can only represent continuous correlation, not the discrete, hidden "regimes" that drive true climate tail risk.

THE ACTUARIAL BLUEPRINT

The Data Generating Process (DGP) Laboratory

Step 1: The Book (500 Properties)



Baseline expected loss ratio increases smoothly toward the coast. Elath. Lognormal insured value (mean \$350k).

Step 2: The Dual Regime

Path A
(Normal Year)

Independent lognormal draws.
No cross-property structure.

Path B
(Systemic Year)

Occurs 1 in 15 years. Triggers a shared, spatially-correlated **RBF random field** $z_{it}(t)$ with length scale 0.5 deg and a fixed severity multiplier of 6.0.

Visual Formula Box

$$L_{i,t} = V_i * \mu_i * m_{sys} * \exp(\epsilon_{i,t} + z_i(t))$$

Individual Property Loss Terms
(Lognormal Model)

Systemic shock mathematically **multiplies loss simultaneously** across the spatial field.

THE ACTUARIAL BLUEPRINT

The 4-Method Actuarial Ladder

1. Independence

Textbook CLT baseline. Fits marginals independently. No correlation at all.

2. Flat Correlation

The industry standard (Solvency II / Rating agencies). One average pairwise correlation number applied via a single-factor model.

3. Vanilla Spatial GP

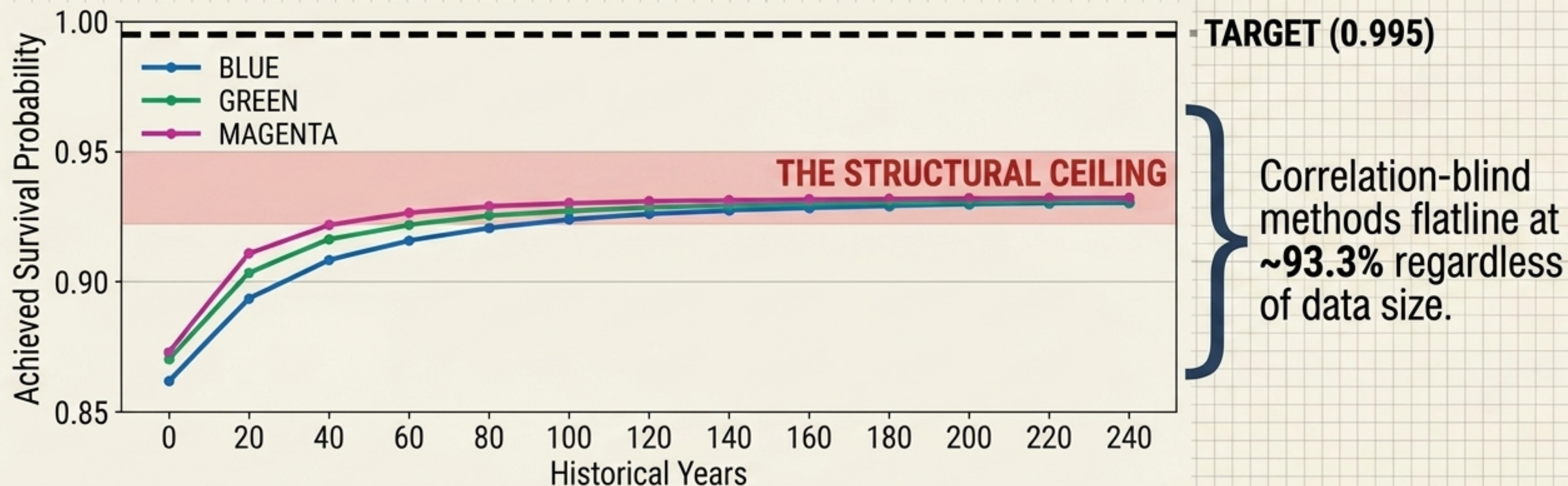
Better shape, real distance decay. Fits a 3-parameter RBF correlation matrix over lat/lon via repeated-measures log marginal likelihood. No regime concept.

4. GP + Regime Mixture

The only model built to represent the DGP. Two GP components mixed by an estimated probability. (Tested as both Hard and Soft classifiers).

THE ACTUARIAL BLUEPRINT

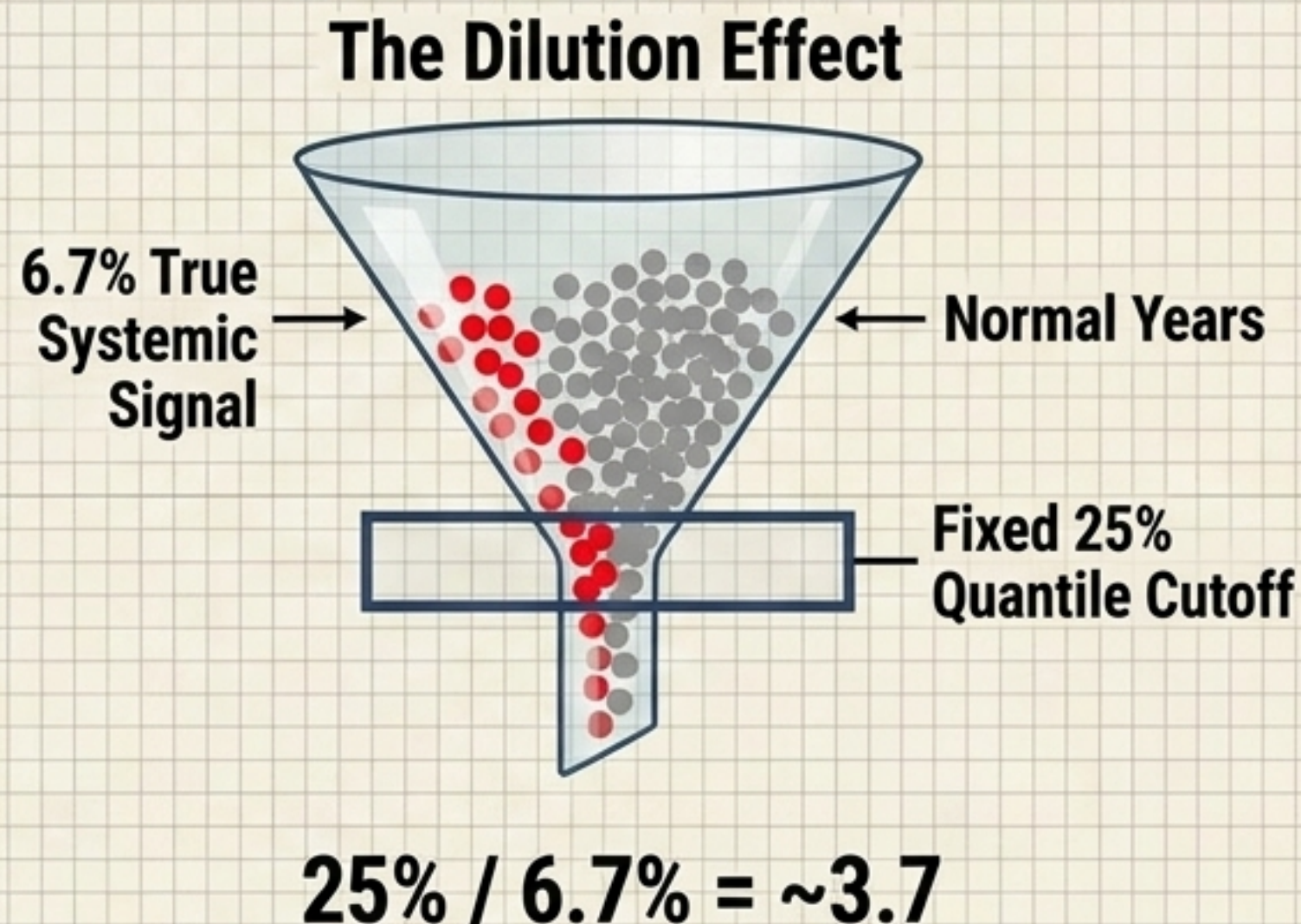
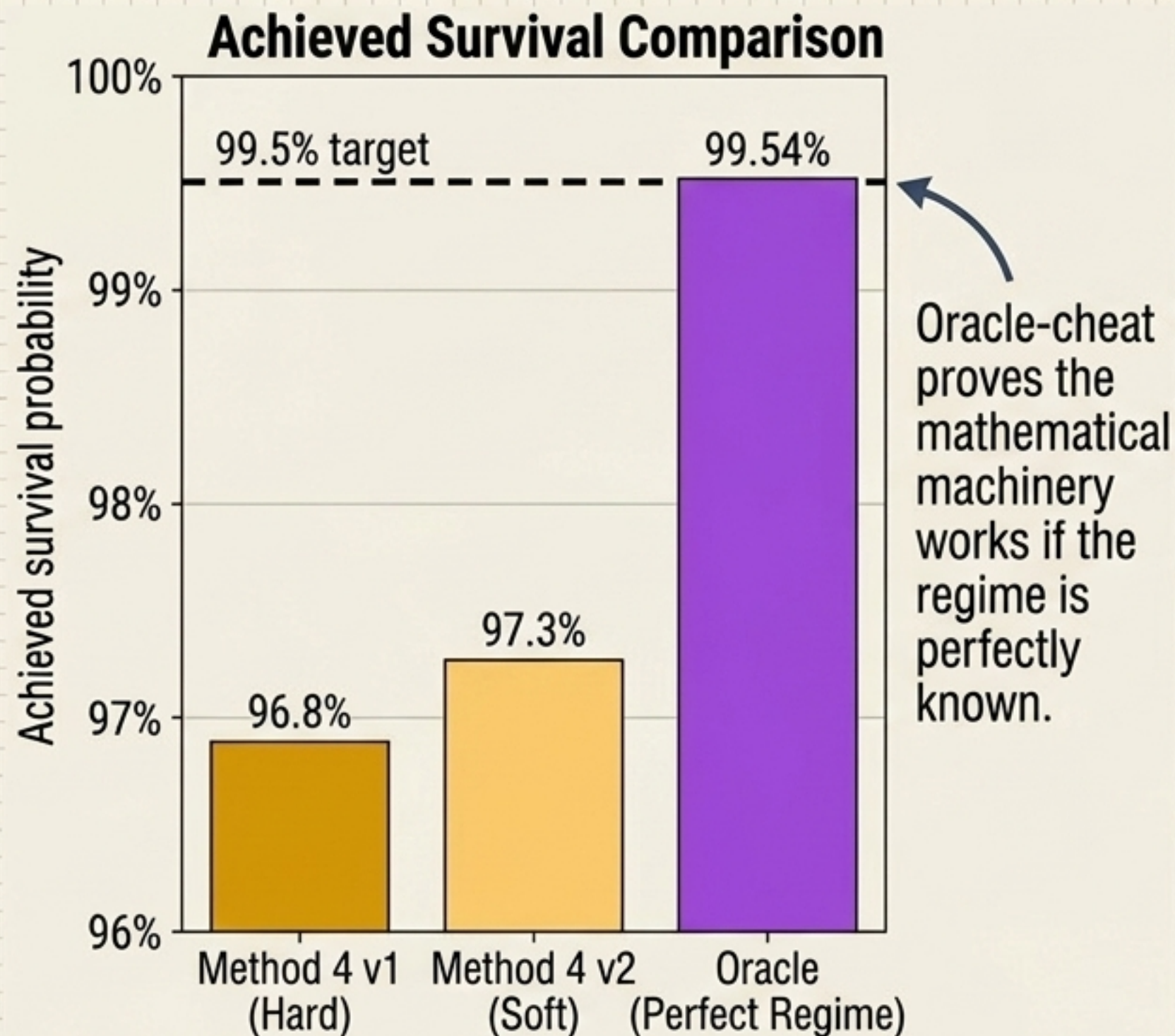
The Data Illusion (The Structural Ceiling)



More historical data does not fix a model that structurally cannot represent the mechanism that matters. Elliptical models are fundamentally blind to tail dependence.

THE ACTUARIAL BLUEPRINT

Why the “Right Model” Fails: The Hard Partition Dilution



To ensure mathematical matrix convergence, the model forces 3.7 normal years in for every 1 genuine systemic year. This mechanically dilutes the fitted systemic severity down, blinding the model to the true extreme tail.

THE ACTUARIAL BLUEPRINT

The Mathematical Breakthrough: Soft EM Classification

Hard Assignment (The Flaw)

$$\{0, 1\}$$

Binary inclusion. A year is either entirely in or entirely out. Discards valuable tail information.

Soft Weighting (The Fix)

$$w_t \text{ in } [0, 1]$$

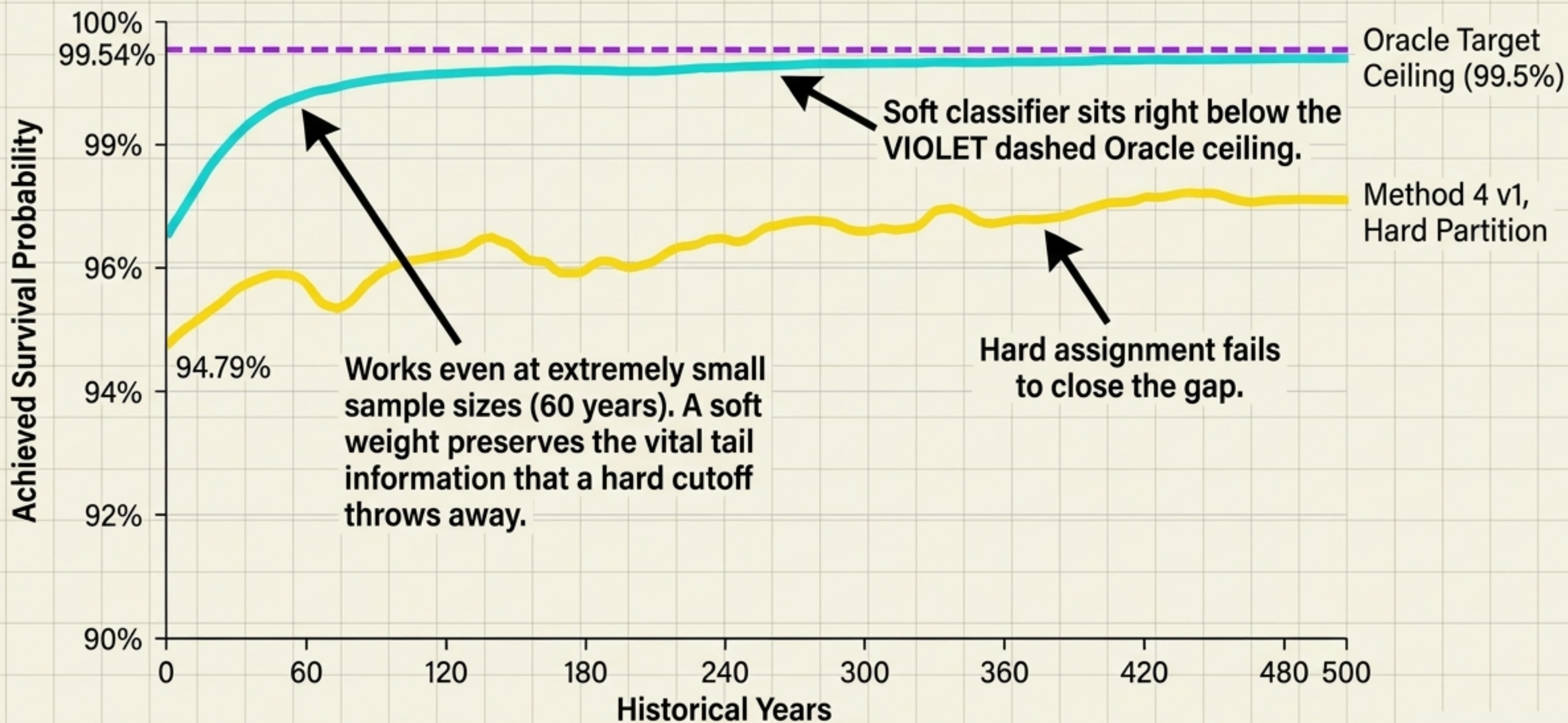
Posterior responsibility for each year from the Gaussian Mixture Model. Every year contributes.

$$\log p(\dots) = -\frac{1}{2} \sum_{t=1}^T [w_t \cdot \mathbf{R}_t^T \cdot \Sigma^{-1} \cdot \mathbf{R}_t]$$

The weighted repeated-measures log marginal likelihood completely eliminates cutoff tuning knobs. Preserves all data, dynamically adapts to scale.

THE ACTUARIAL BLUEPRINT

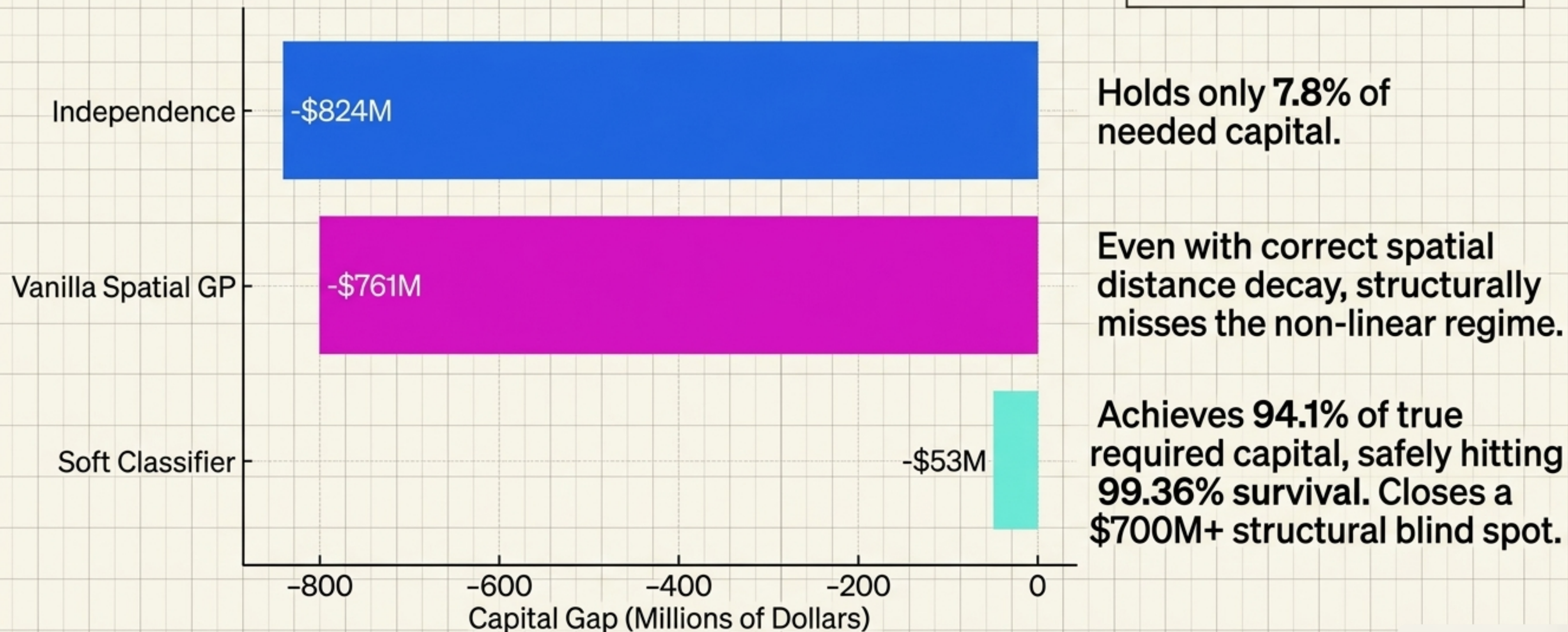
Breaking the Ceiling



THE ACTUARIAL BLUEPRINT

The Dollar Gap at 45,000 Properties (Out-of-Core Scale)

True Oracle Capital Requirement:
\$893 Million

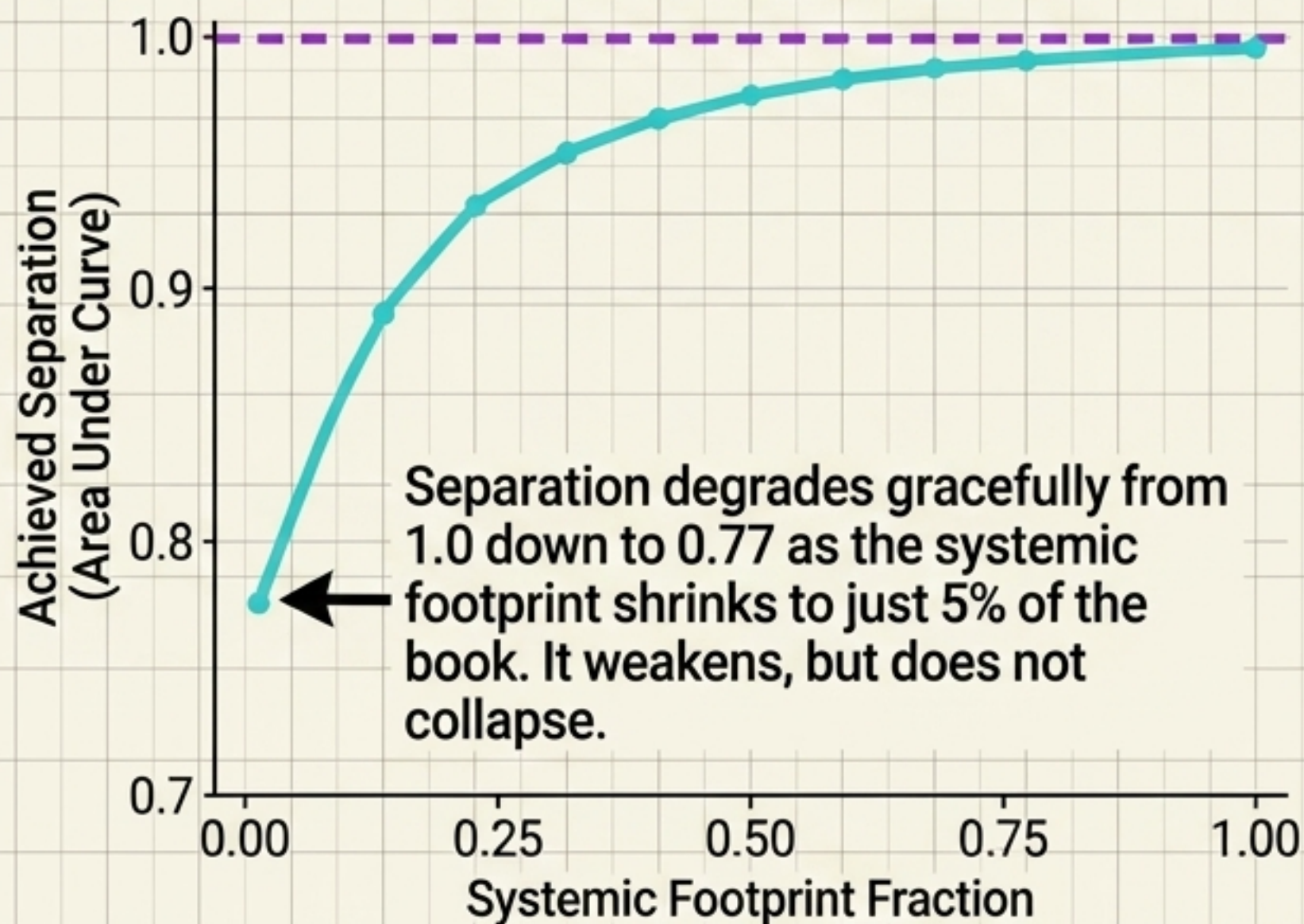


THE ACTUARIAL BLUEPRINT

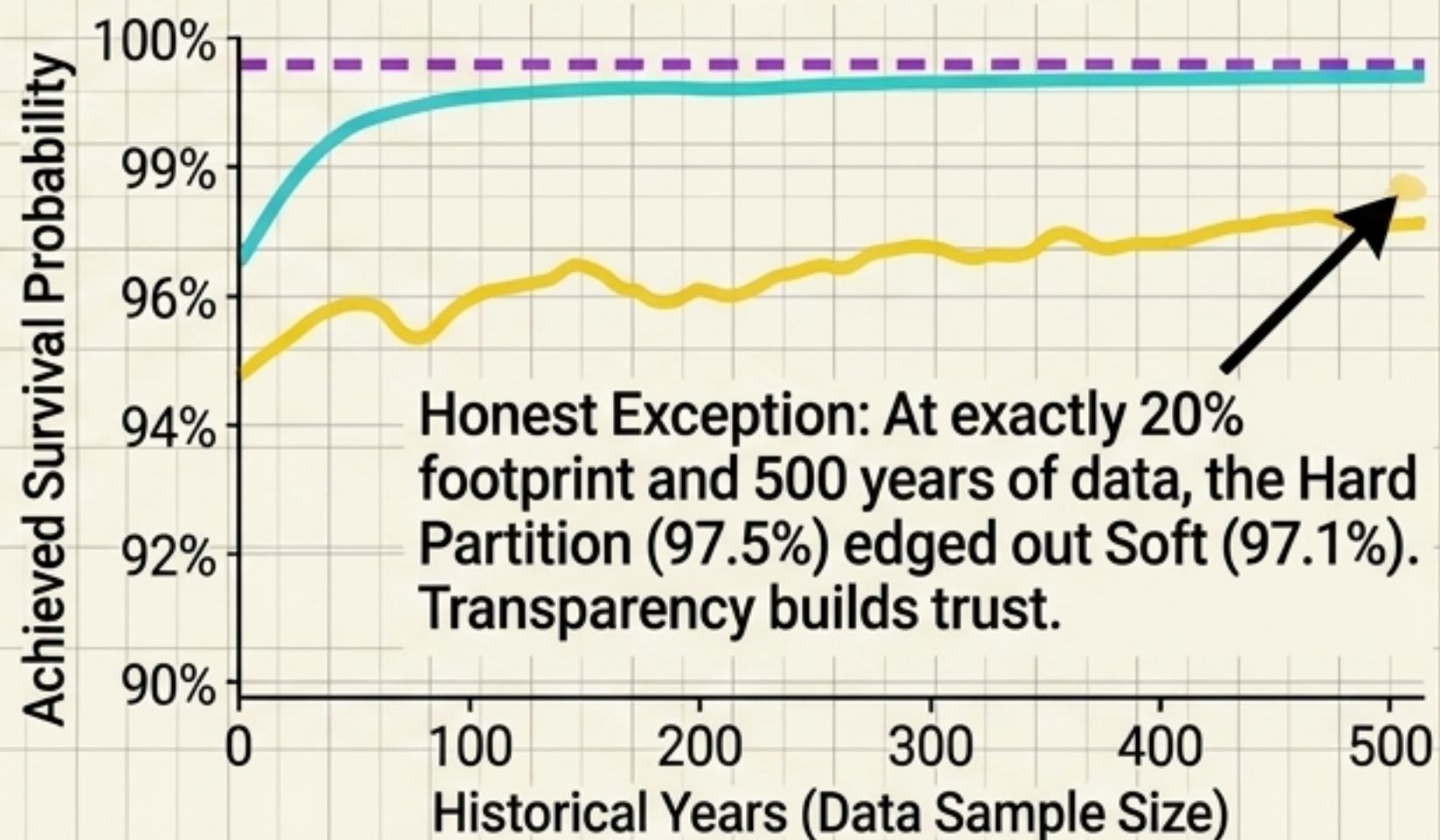
Robustness Test I: The Localized Climate Footprint

What if the systemic shock only hits a small fraction of the book?

Classifier Separation



Achieved Survival



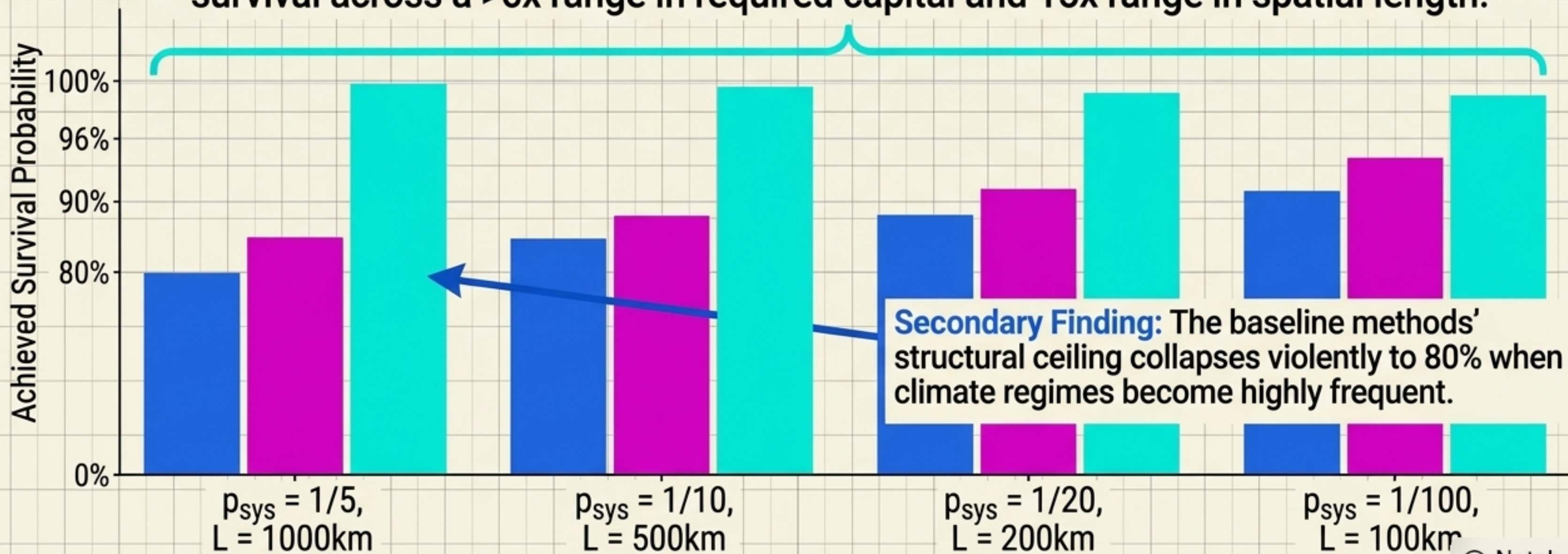
Overall, the Soft Classifier (**Aqua**) maintains its structural edge at every footprint level.

THE ACTUARIAL BLUEPRINT

Robustness Test II: Parameter Sweeps

Achieved Survival Across Parameter Configurations

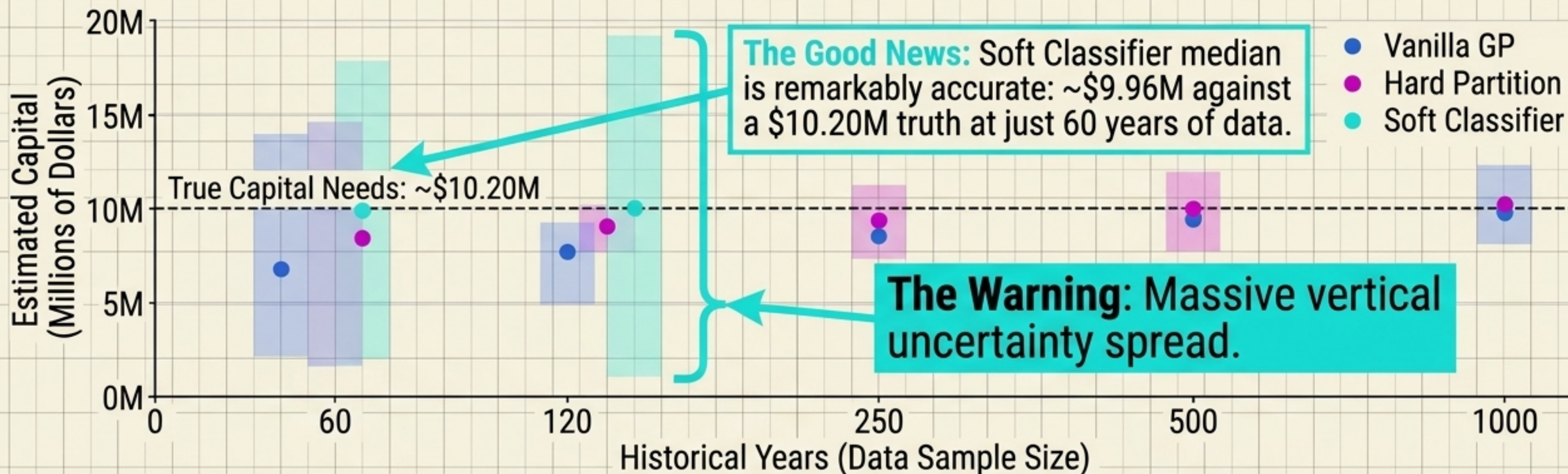
The Constant: Soft Classifier uniformly stays between 98.6% and 99.7% achieved survival across a >6x range in required capital and 15x range in spatial length.



THE ACTUARIAL BLUEPRINT

The Unresolved Caveat: Sampling Uncertainty

Achieved Capital Estimation (Median & Spread) Over Historical Years



CORE TAKEAWAY: Soft classification fixes systematic bias, but sampling uncertainty remains massive. An insurer relying on a single 60-year historical draw could estimate capital anywhere from \$1.5M to \$14.2M. Point estimates are not a substitute for Bayesian uncertainty quantification.

THE ACTUARIAL BLUEPRINT

Synthesis & Industry Imperatives

Architectural mandates to close the \$1.8 Trillion protection gap.

Stop Relying on Shape Alone



Distance-decay and flat correlation models (Gaussian copulas) hit a hard ~93% survival ceiling. They mathematically cannot represent non-linear climate regimes.

Avoid Data Dilution



Hard-partitioning models (binary stress tests) dilute severe tail events by mechanically mixing them with normal years to force matrix convergence.

Implement Soft-EM Spatial Models



Responsibility-weighted repeated-measures marginal likelihoods close the structural gap. They **preserve all data**, dynamically adapt to scale, and successfully **reserve 94%+ of true required capital** on portfolios of 45,000+ properties.